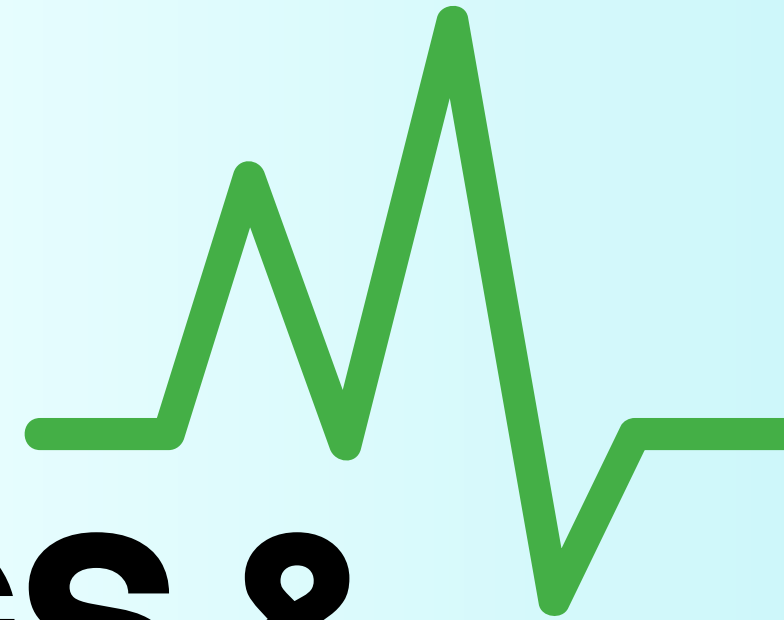


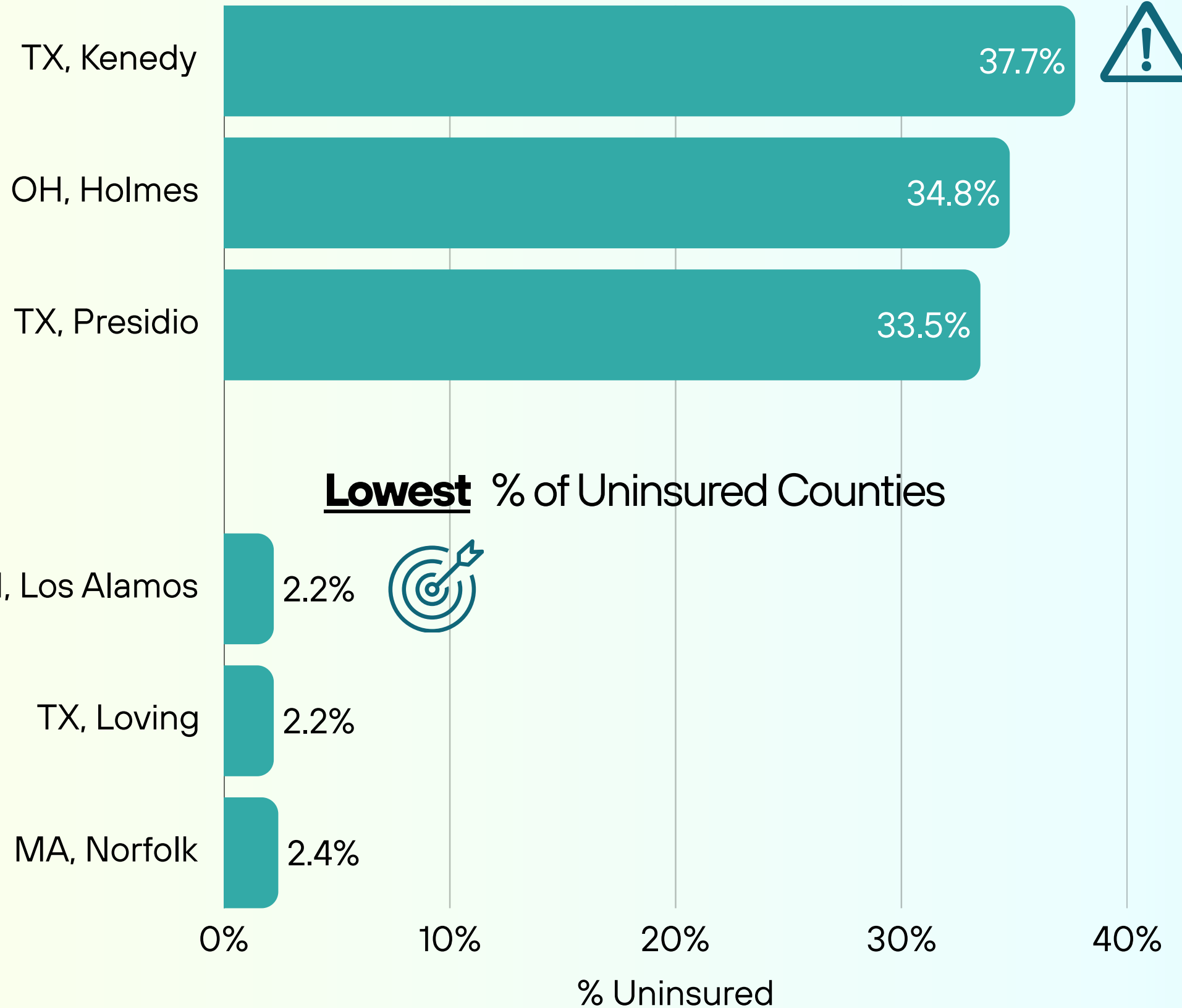
TOWSON UNIVERSITY DATA ANALYTICS COMPETITION 2026

2025 COUNTY HEALTH RANKINGS & ROADMAPS

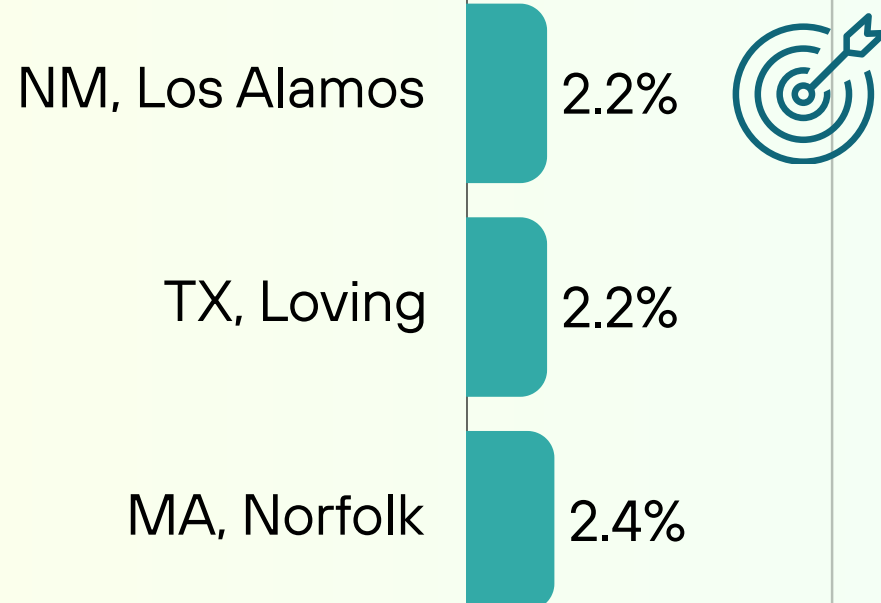


A Statistical Analysis of the CHR&R Dataset: Examining the Correlation
Between Health Insurance Coverage and Population Health Outcomes
Across U.S. Counties.

Highest % of Uninsured Counties



Lowest % of Uninsured Counties



America's Most and Least Insured Counties

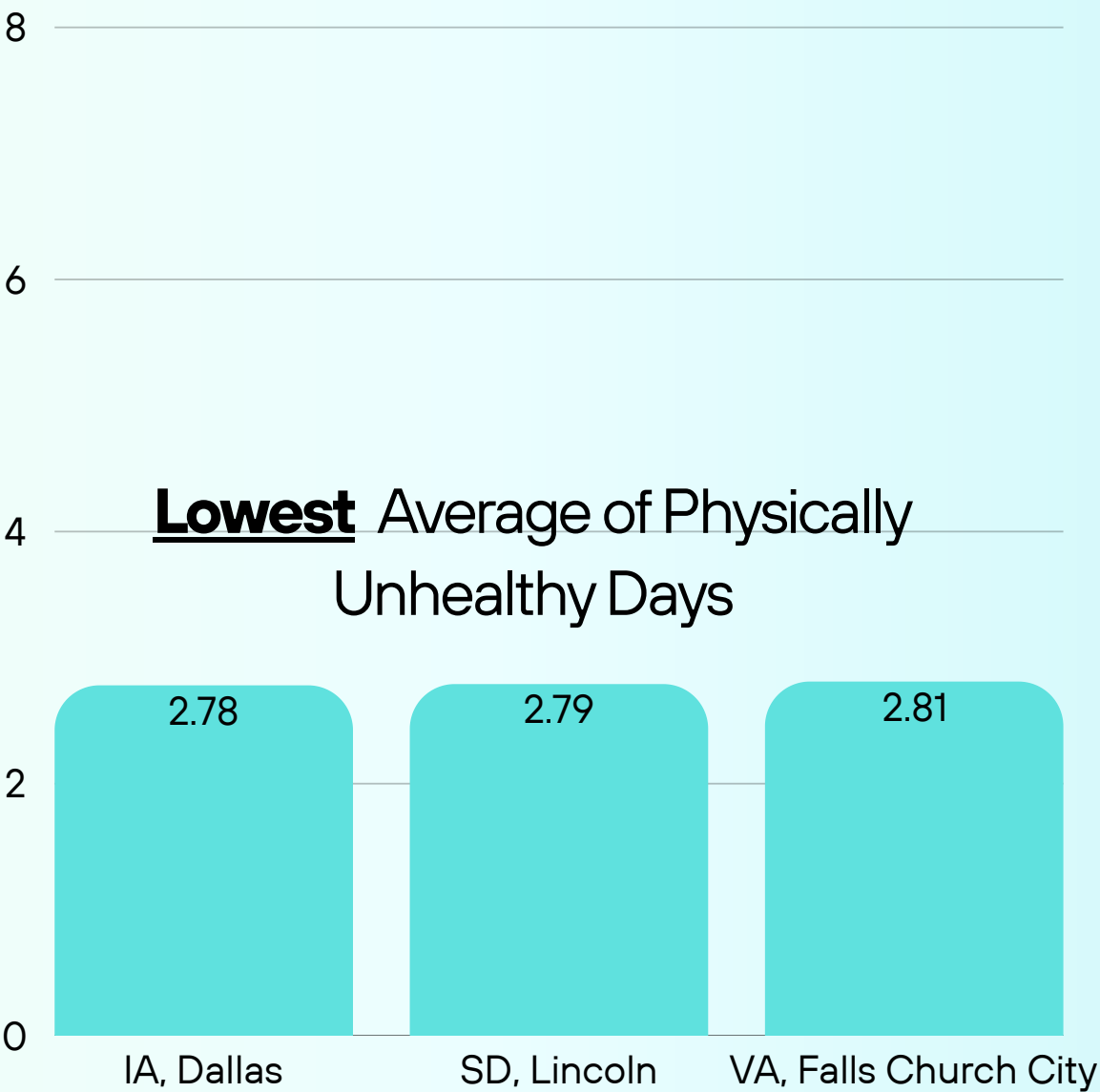
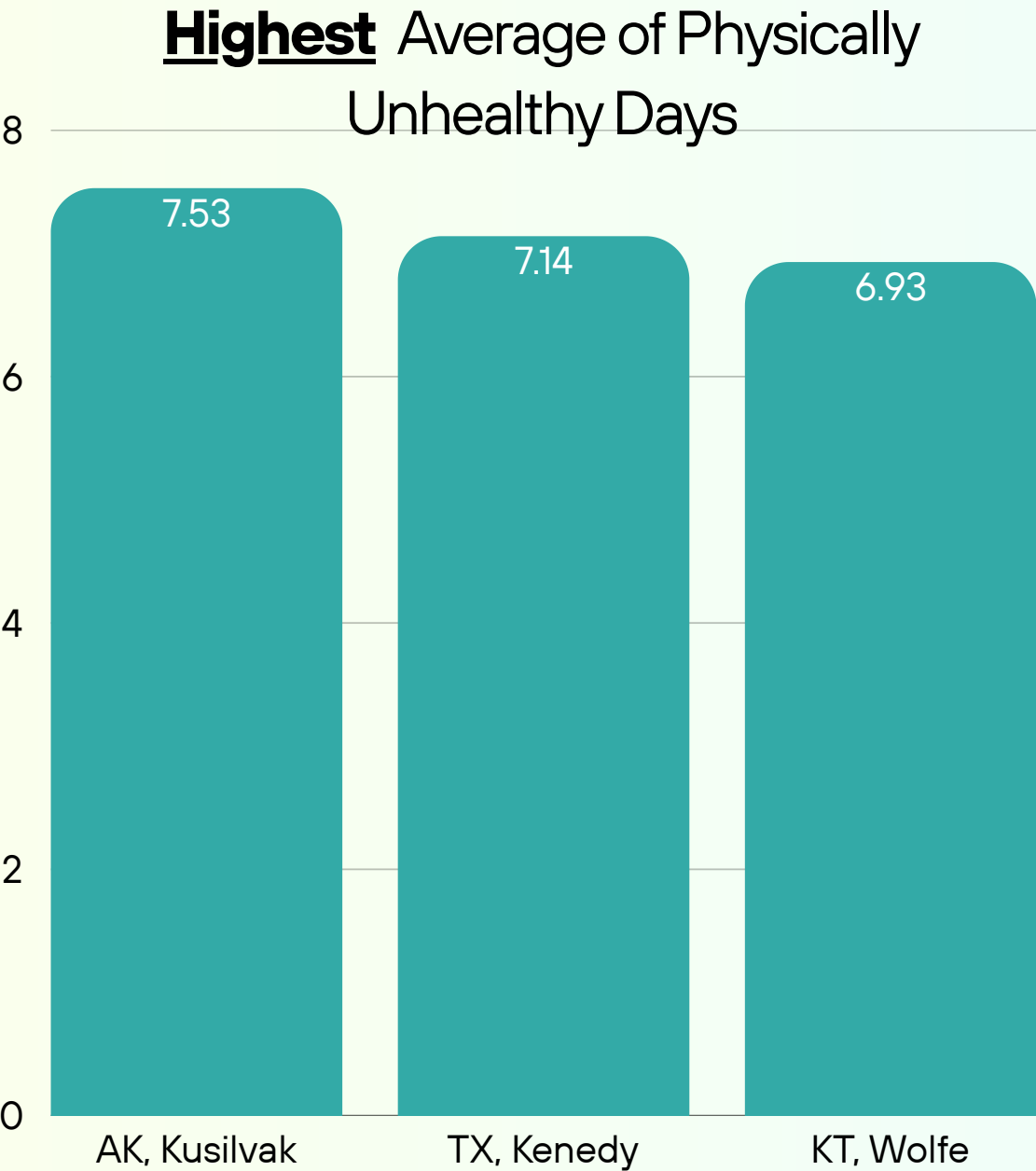
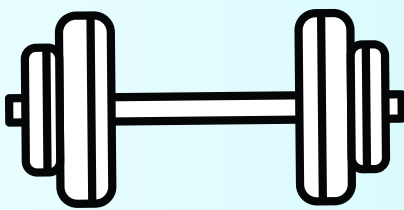
10.3%

UNINSURED AVERAGE PER COUNTY

National Insurance Gap: Kenedy County, TX residents are nearly 18 times more likely to be uninsured than those in Los Alamos, NM—a 35.5-point gap showing major geographic inequality.

- **Population Factor:** Loving County, TX has the smallest U.S. population, making its 2.2% uninsured rate a statistical outlier. Small populations often create extreme highs or lows.

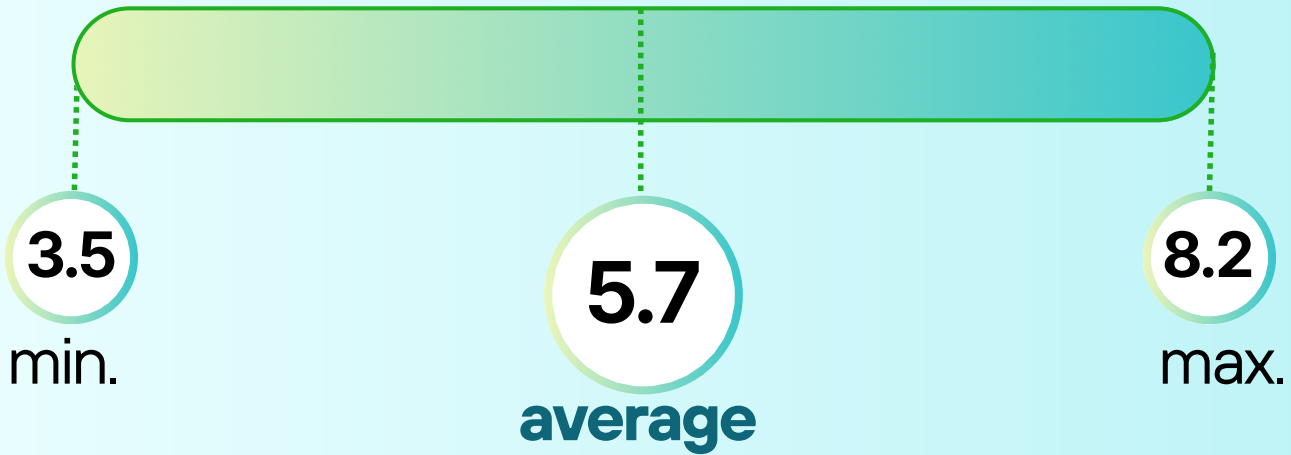
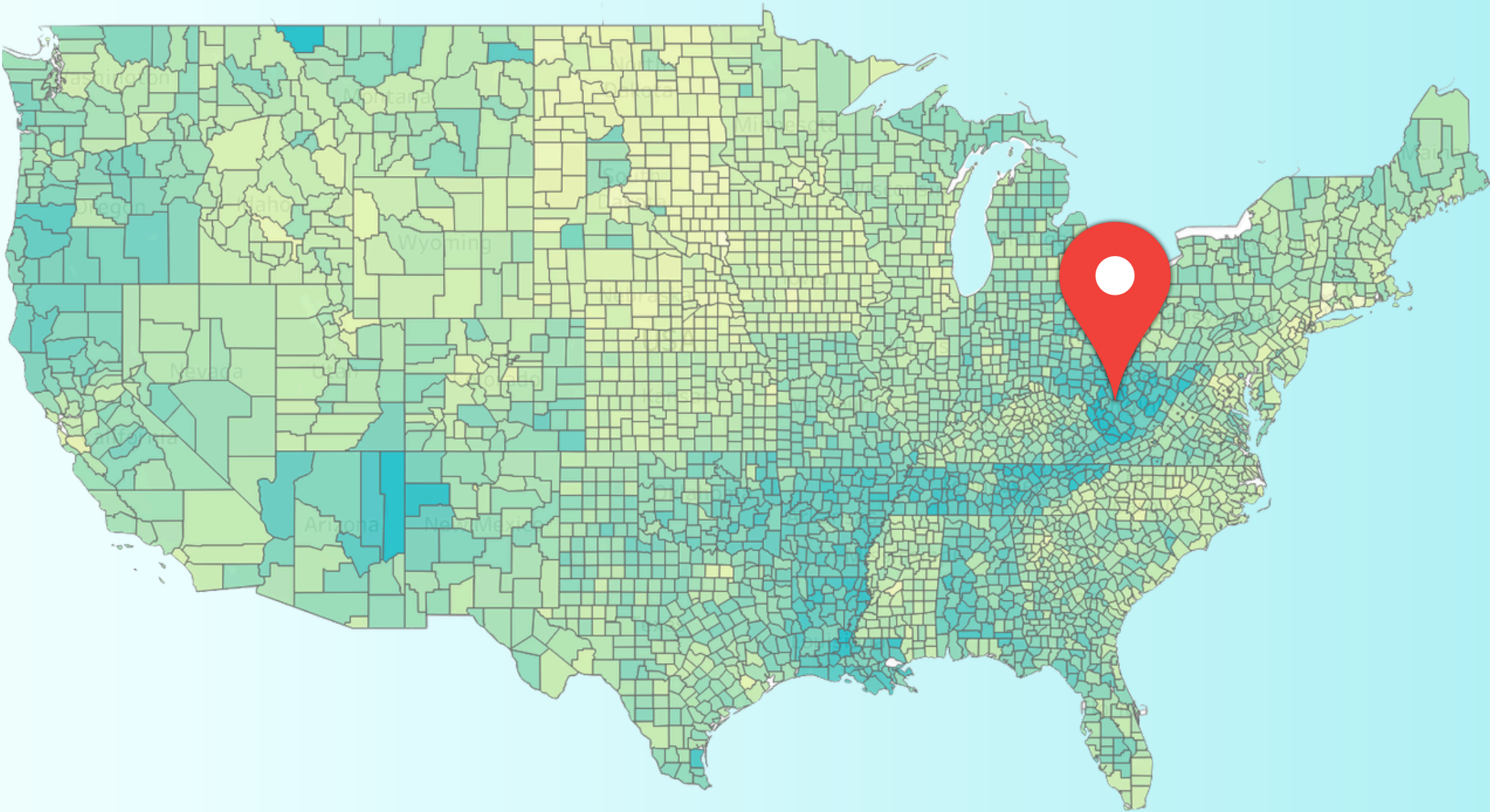
The 5-Day Divide: Extremes of Physical Health in America



- **Kenedy County, TX:** Averages 7.5 unhealthy days per month, correlating with its high uninsured rate.
- **Top Performers:** Dallas (IA), Lincoln (SD), and Falls Church City (VA) report the lowest levels, averaging under 3 days.

West Virginia vs. The Midwest: A 5-Day Gap in Wellbeing

Geographic Distribution of Mental Well-being



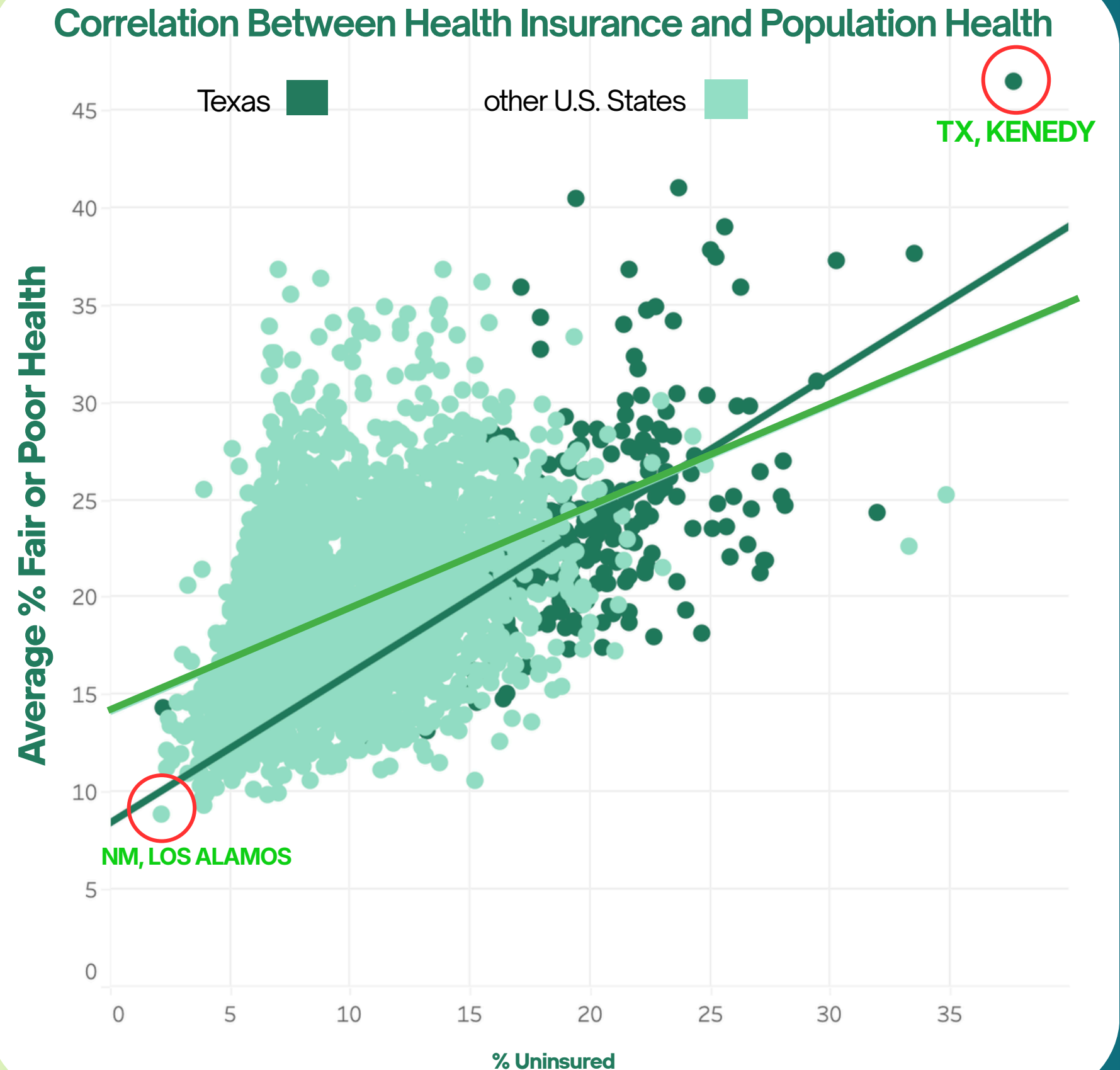
Category	County & State	Avg. Days
Highest	Mingo, WV	8.24
	Logan, WV	7.77
	McDowell, WV	7.69
Lowest		
	Lincoln, SD	3.5
	Carver, MN	3.91
	Clark, SD	4.06

- **WV Clusters:** Mingo, Logan, and McDowell (WV) lead with the highest averages (7–8 days/mo).
- **Midwest Leaders:** Lincoln (SD), Carver (MN), and Clark (SD) maintain the lowest averages (3.5–4 days/mo).
- **Comparison: A nearly 5-day difference exists between the top and bottom-tier counties.**

Texas as a National Outlier: The High Cost of the Uninsured

- **National Extremes:** Texas accounts for the three lowest-ranked counties for general health in the U.S. (Kenedy, Zapata, and Dimmit).
- **The Disparity:** Kenedy County (46.5%) reports nearly 5 times the rate of poor health seen in top-performing counties like Los Alamos, NM (8.8%).

Texas counties (dark green) dominate the "**High Uninsured**" sector of the graph. This establishes a direct statistical link between a lack of health insurance and a significantly higher percentage of the population reporting Fair or Poor Health



National Health Landscape: % of Fair to Poor Health Status



Appalachia: West Virginia remains a major dark spot, specifically **Mingo and Logan** counties, which lead the U.S. in mentally unhealthy days.



min.

19.5%

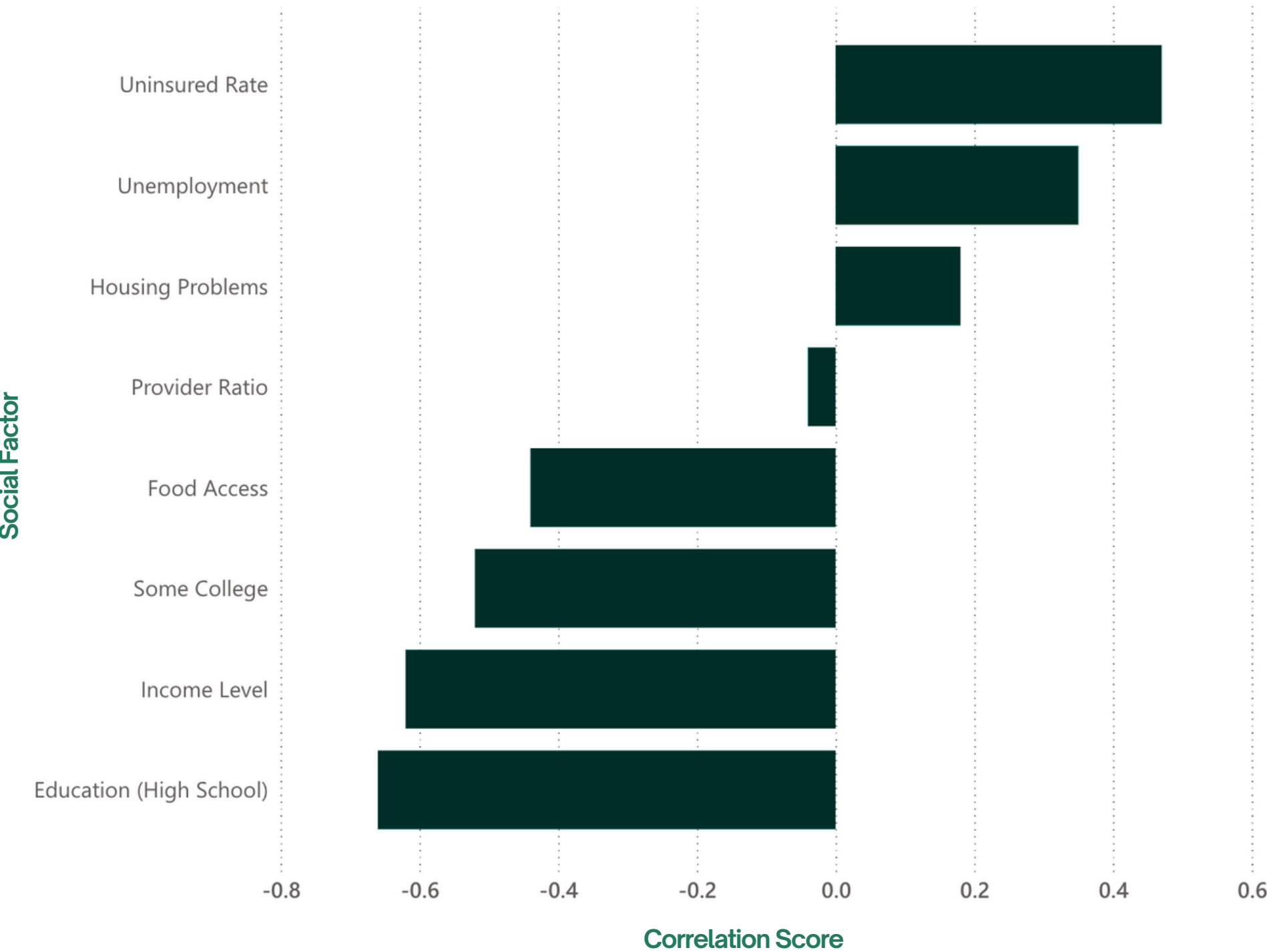
average

46.5%

max.

Social Drivers of Mental Health Disparity in West Virginia

Correlation Score by Factor



Factor	Influence on Mental Health	Direction
HS Graduation	Very Strong	↑ Education = ↓ Unhealthy Days
Income Level	Strong	↑ Income = ↓ Unhealthy Days
Some College	Strong	↑ Higher Education = ↓ Unhealthy Days

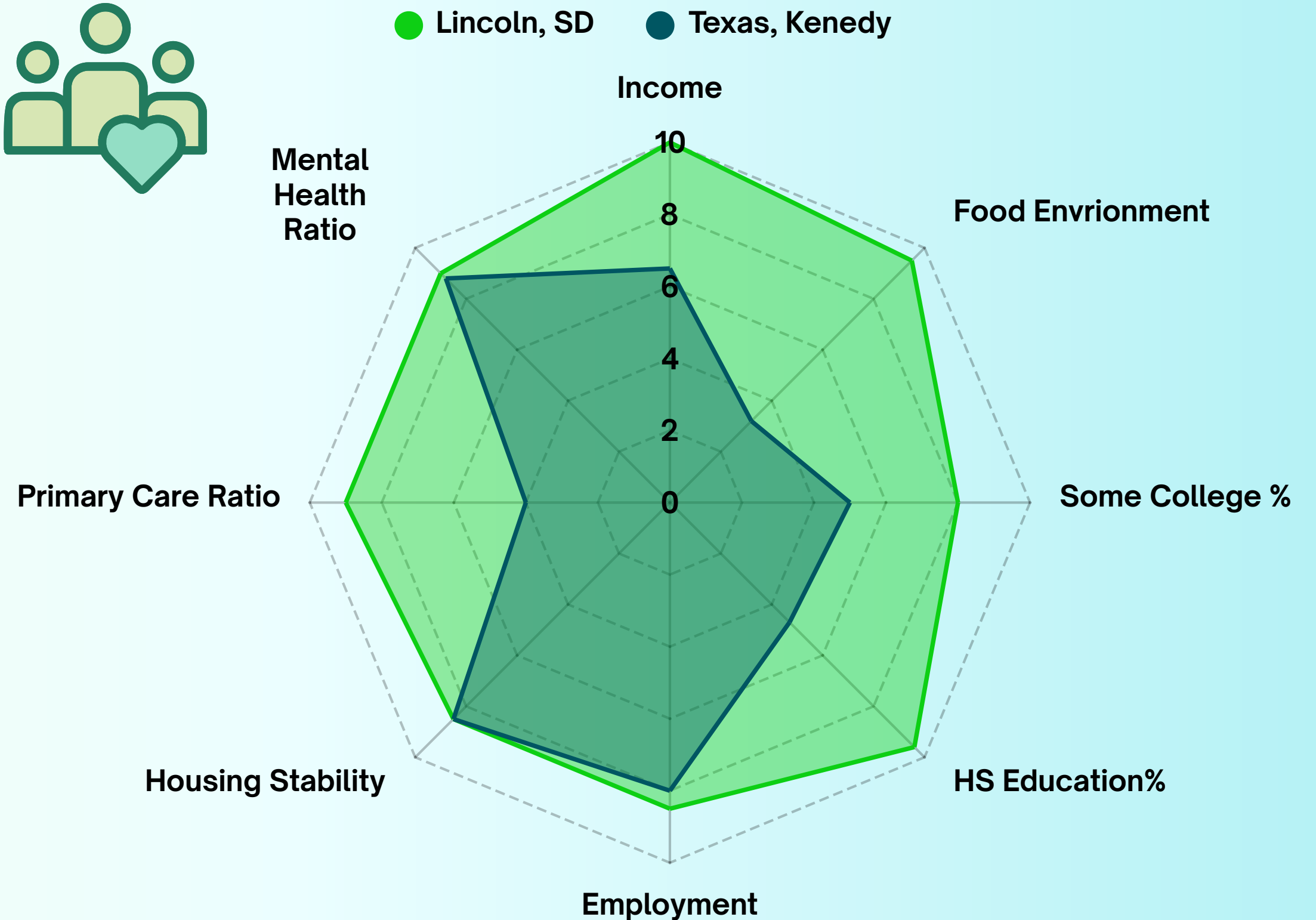
The Health Equity Web: Lincoln, SD vs. Kenedy, TX

The data is clear: fragile foundations in education and food access lead directly to poorer health outcomes.



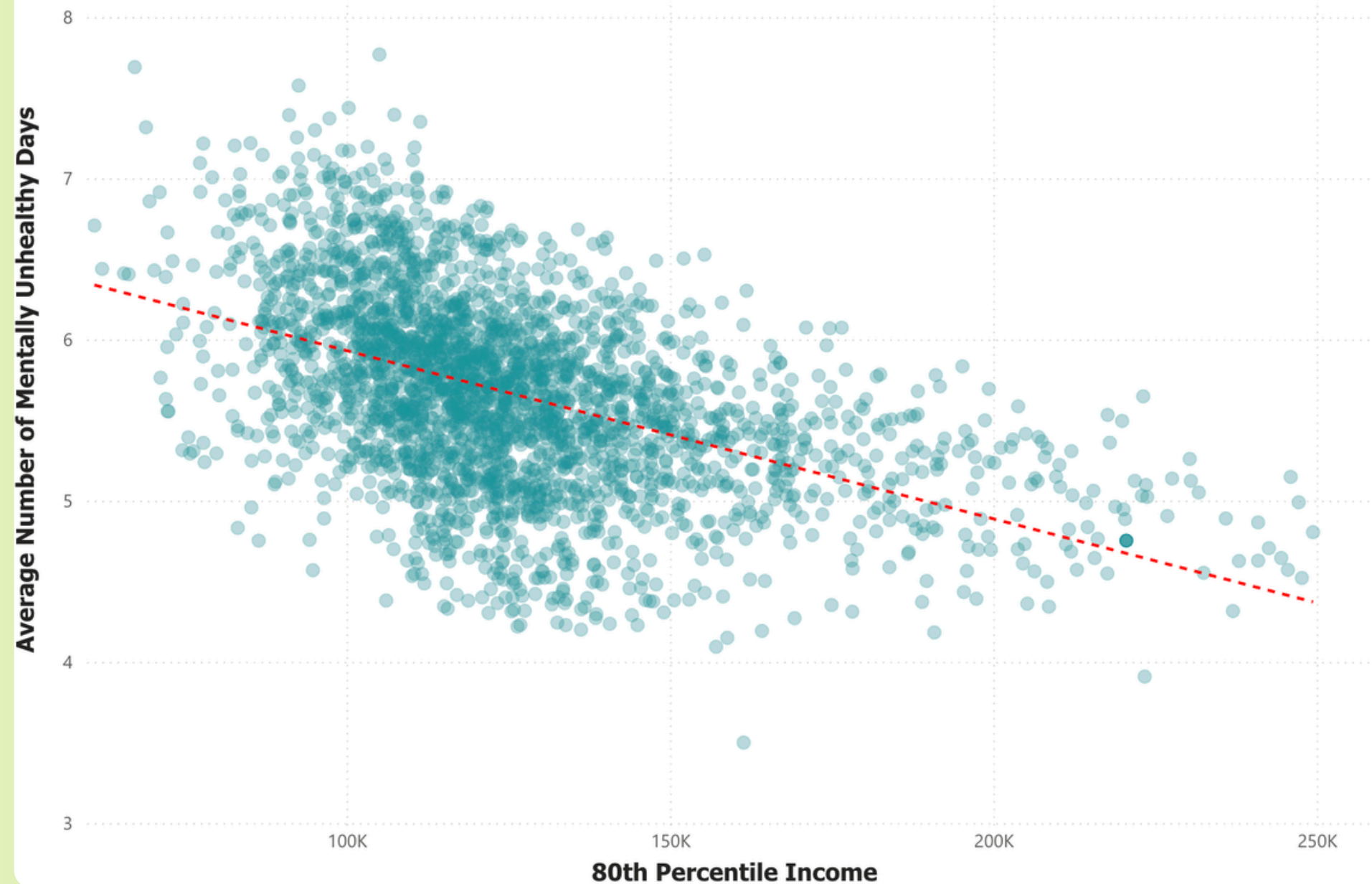
Key Insight: While both counties show strong clinical provider ratios, the large difference in their final health outcomes is driven by social factors like education and food.

Social Factors, not just medical access, are the primary drivers of the national health divide



Scores are normalized on a 1-10 scale to compare diverse variables

Economic Security vs. Mental Wellness



Income as a Predictive Factor of Mental Well-Being

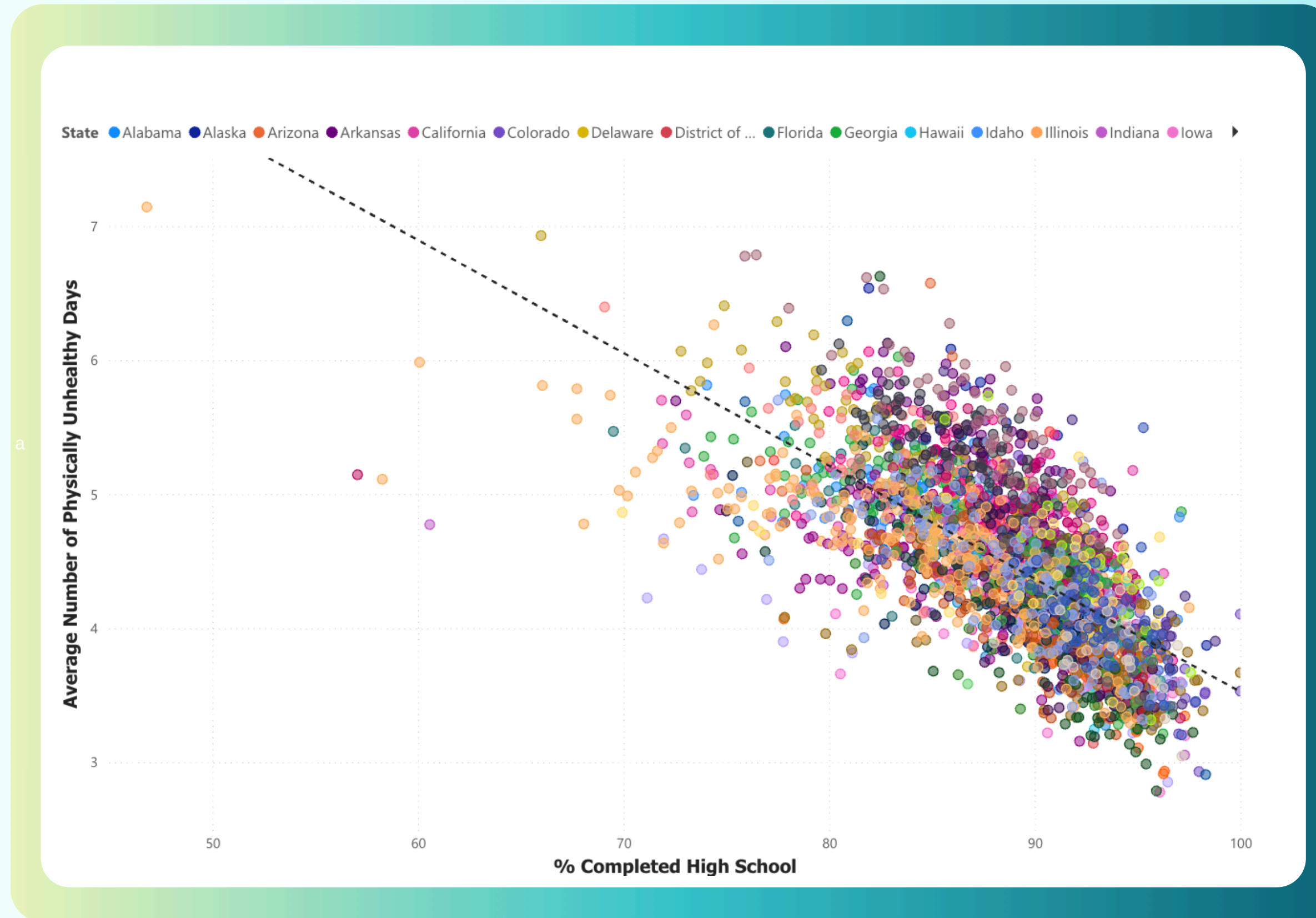
A strong negative correlation ($r = -0.47$) indicates that higher income is the leading predictor of better mental health outcomes.

- This Income Factor is followed closely by Education and Food Security, proving that mental wellness is deeply rooted in a community's financial foundation.

Education as a Predictive Factor of Physical Well-Being

High school completion is the single strongest shield against physical illness. The powerful negative correlation ($r = -0.68$) demonstrates that as graduation rates rise, physical distress significantly declines

- This Education Factor is closely followed by Income and Food Security, emphasizing the importance of social factors again



Counties to **Focus** on:

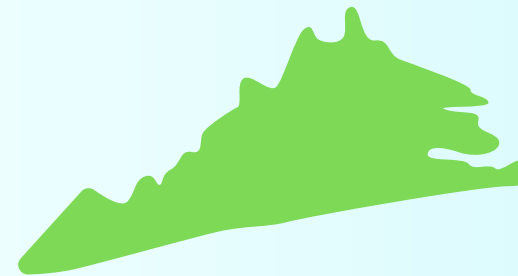


West Virginia: Mingo County, Logan County, and McDowell County consistently face a large number of mentally unhealthy days. West Virginia must prioritize economic improvement and mental health infrastructure.



TX, Kenedy: Kenedy County consistently ranks among the lowest in national health outcomes. To bridge this gap, policy interventions must prioritize educational infrastructure and food security.

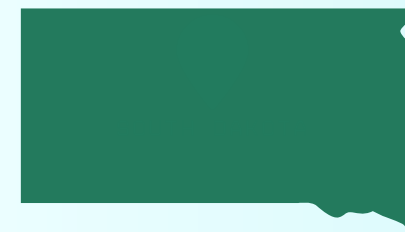
Aspirant Counties:



VA, Falls Church City: Falls Church achieves the dataset's highest graduation rate, directly influencing its rank as the 3rd healthiest city for physical well-being and strong economic stability.



NM, Los Alamos: Los Alamos achieves near-perfect high school completion rates, directly correlating to its standing as the nation's leader in physical resilience with the lowest number of physically unhealthy days in the dataset



SD, Lincoln: Lincoln excels across all key health factors identified. With top-tier scores in education and income, it maintains a strong foundation due to high food security and minimal unemployment.

Other Factors to Consider:



population

Extremely low populations can skew results and create statistical outliers. Loving, Texas, appears as a national leader in insurance coverage (2.2%) primarily because it is the smallest county in America, where minor shifts in individual data points impact the overall average.



age

A county's median age is a primary confounding factor in health data. Older populations naturally report a higher frequency of physically unhealthy days regardless of income or education, meaning we must "normalize" age data to see the true impact of social determinants.



digital accessibility

High-risk counties often suffer from a lack of digital resources and internet. This prevents access to telehealth services and remote economic opportunities, further widening the health divide

Summary:



Data identifies Education and Income as the leading predictors of health outcomes, supported by the critical factors of Food Security and Housing Stability. To bridge the health gap in counties like Kenedy and Mingo, we must move beyond clinical treatment and prioritize this social infrastructure as the true 'medicine' for long-term community improvement.